

Detecting Genuine Price Anomalies for Battery Storage Arbitrage in CAISO: A Multi-Model Benchmark and Adversarial Assessment

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Abstract—The main source of revenue for the BESS lies in a few price spikes per year. However, most previous studies examine the ability of anomaly detectors to operate on historical data with artificial price spikes. Such methodology is reasonable in the case of cybersecurity research. Nevertheless, it does not reflect the reality of p2p trading – the price spike is a source of profit for the parties involved, not corrupted data that needs to be removed. In our further research paper, we are testing the ability of unsupervised anomaly detectors to detect profitable price events using three years of raw California ISO day-ahead price data (from 2023 to 2025, without any injections or synthetic data). Under such conditions, recurrent models emerge as the winners. LSTM autoencoder achieves an ensemble AUROC score of 0.859 (0.921 when tested on a seasonal rolling window), and GRU autoencoder surpasses the competitor, achieving an AUROC score of 0.882 and training 25% faster than the counterpart. With the help of such an anomaly-detection signal, we can increase the arbitrage profits of the tabular Q-learning policy by 4x, to \$11,957 over six months. Another area of interest was stress testing the system to defend against adversarial manipulation; threshold-based policies eventually incur losses because they generate false positives and discard the most profitable time periods, requiring a profitability precision of more than 100%. However, a classifier trained to detect adversarial attacks can achieve 33.4% precision, achieve 89.1% accuracy, and mitigate 67.2% of the losses incurred by the attack.

Index Terms—Adversarial robustness, anomaly detection, battery energy storage systems (BESS), CAISO, energy arbitrage, graph neural networks, reinforcement learning, time-series forecasting.

I. Introduction

Electricity prices in California have grown considerably more unpredictable in recent years. Solar output during the afternoon sun usually pushes locational marginal prices (LMPs) down toward zero, and sometimes below it, only for prices to snap back once the sun sets and gas and coal plants jack up prices to cover the evening peak. BESS exploits this daily rhythm, charging their systems while power is cheap and discharging as prices climb.

Most days, these swings follow a pattern dictated by the work schedule and the weather. However, a battery’s annual profits tend to depend on a handful of extreme hours. Hourly prices can soar over \$1,000/MWh during heatwaves, transmission outages, and sudden generator trips, and a 10 MW/10 MWh battery that discharges just

once during such a spike can earn about as much as it would have made in a whole week otherwise. It is therefore a high priority for storage operators to detect these rare hours as soon as possible or to anticipate them.

Previous SOTA models have mostly treated price anomaly detection as a cybersecurity problem [1], [2]. This approach begins by using historical data without anomalies, introduces artificial price spikes, and checks whether the detector can detect them. This framework can be useful for protecting against cyber-attacks. However, it cannot demonstrate the real trading process, the sources of prices, or the legal mechanisms of market settlement that define them. If the detector flags a valid \$1,000/MWh price spike as an anomaly and prevents battery discharge, the operator will incur significant financial losses.

Such a gap explains why, in the main experimental sections III-VI, we use only the actual market prices, obtaining settlement information from CAISO’s OASIS [3] website without any synthetic additions. In Section VII, we investigate the second question, which arises together with the problem stated above. Almost all SOTA approaches consider price anomaly detection as a cybersecurity problem [1], [2]. They start with the initial clean data, add artificial spikes, and see whether any detectors detect them as anomalies. Such an approach helps defend against cybersecurity attacks; however, it does not capture the real trading dynamics, the sources of prices, and the impact of the legally binding market settlement system on their formation. In such a case, if the detector flags the actual \$1,000/MWh spike as an anomaly and prevents battery discharge, the operator might incur significant losses. Does the attacker really inject fake spikes into this data flow?

Using historical market data has two pragmatic issues. The first is that the historical prices provided by market operators cannot be taken as ground-truth labels; therefore, the model needs to be validated for statistical consistency and, ultimately, profitability. The second and even more important issue is that mistakes are not symmetrical in any sense. While a mistake in image or text recognition is just an annoyance, a mistake that prevents the battery from discharging into a real \$1,000/MWh spike costs money.

This study contributes in three ways:

- 1) Benchmarking anomaly detectors. We benchmark Isolation Forest, a VAE, a denoising diffusion model, single-node LSTM and GRU autoencoders, a multi-

variate **LSTM**, and a graph attention network over 25,944 hours of real **CAISO** data, five random seeds, and two different test windows.

- 2) **Economic battery mapping**. Mapping of detector readings into the economics of battery is done for dispatch strategies using rules-based, directional, and reinforcement learning approaches. It turns out that tabular Q-learning yields the greatest benefits, with a nearly fourfold increase in profit.
- 3) **Economic break-even precision metric**. We develop an economic breakeven-precision metric, p^* , for adversarial feed tampering, and demonstrate why simple threshold rules are unprofitable even as a trained classifier breaks even at a comparatively low 33.4% precision, and retrieves most of the lost returns.

II. Related Work

This work sits at the intersection of electricity price forecasting [4], unsupervised time-series anomaly detection [2], [5], generative modeling [6], [7], and battery dispatch optimization [8]–[10].

Electricity Prices and Anomaly Detection. Traditional methods for detecting price anomalies relied extensively on rolling statistics, moving averages, and local z-scores. While these techniques are easy to compute, they are inherently flawed because fixed thresholds cannot adapt to changing seasons; for example, a threshold calibrated for low winter loads will always yield false alarms during summer afternoon loads.

The deep models learn the usual 24-hour price pattern directly from historical data; there are no predefined patterns in our models. The recurrent autoencoders (LSTM, GRU) transfer the sequence states forward through time and compress each daily path to a small-size bottleneck and detect the moments when the error of reconstruction becomes large [5], [11]. On the other hand, the variational autoencoders encode price windows into a continuous latent space [6], and the denoising diffusion probabilistic models add Gaussian noise to the input and learn to un-noise this process [7]. Large price spikes are unusual and harder to un-noise, thus leading to large errors.

Battery Dispatch and Trading. For storage scheduling, commonly used methods include MPC [8], linear programming, and dynamic programming [9], while reinforcement learning is increasingly seen as an alternative to these methods [10]. One of the flaws frequently found in benchmarks reported in the literature is an implicit assumption of clairvoyance about tomorrow’s prices; when the optimizer is aware of the prices in advance, then there is no need for anomaly detection at all. This problem is avoided by ensuring that every dispatch instance is performed in a purely causal manner.

III. Data and Chronological Partitioning

We have 3 years of hourly day-ahead LMPs from CAISO from January 1, 2023, to December 31, 2025. We collected

TABLE I
Nomenclature and Parameter Settings

Symbol	Description
s_t	Anomaly score for the 24-h window ending at hour t
H	Window length ($H = 24$ hours)
$x_{t,h}, \hat{x}_{t,h}$	Normalized observed and reconstructed price at hour h
$Q(s, a)$	Action-value estimate for state s and action a
α, γ	Q-learning rate ($\alpha = 0.15$) and discount factor ($\gamma = 0.95$)
r_t	Net arbitrage cash return at hour t (\$)
SOC_t	Battery energy level at hour t ($0 \leq SOC_t \leq 10$ MWh)
$C, P_{\max}$	Rated energy capacity (10 MWh) and power limit (10 MW)
e, dt	Round-trip cycle efficiency ($e = 0.80$) and interval ($dt = 1$ h)
$c_{\text{fee}}, c_{\text{deg}}$	Market fee (\$0.50/MWh) and cell degradation (\$2.50/MWh)

price data from the three primary trading hubs, SP15 (south), NP15 (north), and ZP26 (central California), and four ancillary services (Regulation Up, Regulation Down, Spinning Reserves, and Non-Spinning Reserves) for a total of 25,944 hourly observations. The additional price streams serve as auxiliary contextual information for multi-channel anomaly detection, reflecting the overall tightness of operating reserves. At the same time, the economic trading calculations only evaluate day-ahead energy arbitrage.

Prices reached extremes during this window: SP15 spiked to \$1,143.78/MWh on August 17, 2023, amid a regional heatwave, and dropped to $-\$41.33/\text{MWh}$ on May 19, 2024, when spring solar production flooded the grid.

For a more realistic evaluation of our algorithm’s effectiveness, we break down the time period into three periods without any overlap among them; these include the training period between January 12, 2023 and December 31, 2024 (17,272 hours, approximately 23.7 months), validation period between January 1 and June 30, 2025 (4,272 hours, 5.9 months – used only for threshold calibration), and testing period between July 1 and December 31, 2025 (4,400 hours, 6 months, evaluated once). We carried out a six-month staggered test period between January 1 and June 30, 2025.

Since there are no ground-truth labels for anomalies in historical settlement prices, we generated an objective flag solely for evaluation purposes. An hour is flagged if the price is more than 4 MAD from the 30-day rolling median or in the top/bottom 0.5% of all historical prices (this occurs in 517 out of 25,944 hours, or 2.0%). Models do not have access to this flag at any stage of the training.

IV. Detection Pipeline and Model Formulations

The detection pipeline (Fig. 1) uses a two-level filtering approach: Isolation Forest is used to isolate and remove easy-to-detect outliers, followed by analysis of the remaining possible hours using a deep-sequence model. Nomenclature is presented in Table I.

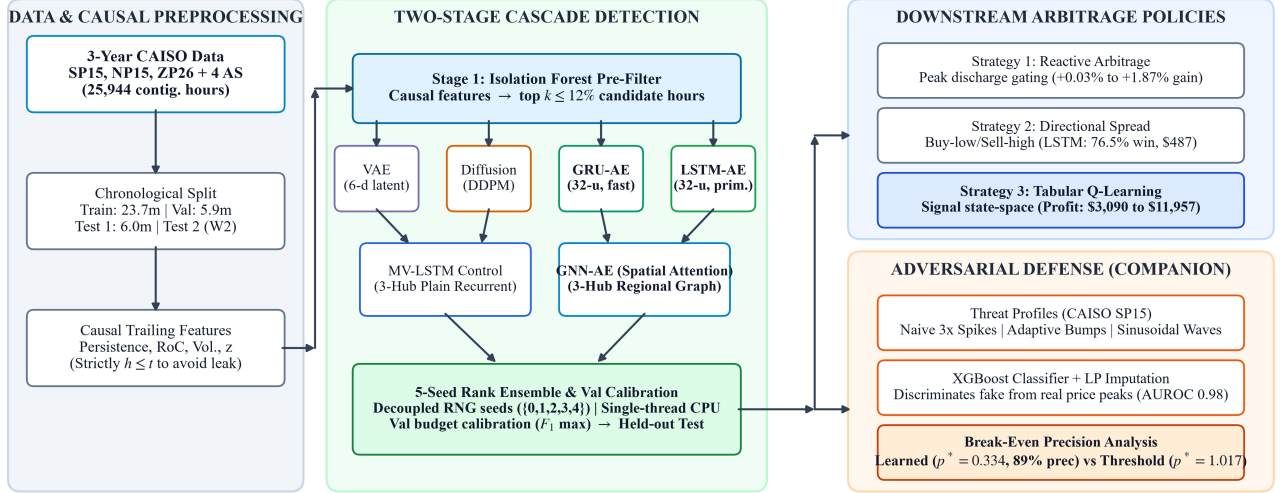


Fig. 1. System architecture. Raw CAISO price feeds move through causal feature extraction into Stage 1 (an iForest pre-filter) and Stage 2 (deep autoencoders and multi-node models); a five-seed ensemble then feeds both the battery dispatch policies and the adversarial defense filters.

Isolation Forest [12] uses nine historical-only features: price rolling slope, non-reversion ratio, hourly diffs, trailing z-scores, volatility, and cyclic time-of-day harmonics.

The candidate windows then go through the deep sequence model, which takes 24-hour normalized sliding price vectors and assesses their anomaly score based on mean squared reconstruction error:

$$s_t = \frac{1}{H} \sum_{h=1}^H (x_{t,h} - \hat{x}_{t,h})^2. \quad (1)$$

Usually, the autoencoder successfully reconstructs the price graph for the particular day. However, when a sudden increase in prices happens, the compressed representation becomes unable to encode the sudden change and s_t

Three models have been implemented in total. VAE has been trained for 40 epochs with the Adam optimizer ($\text{lr} = 10^{-3}$, batch size=64). The VAE architecture consists of fully connected networks with 48 neurons, and the bottleneck is modeled as an 8-dimensional Gaussian distribution [6]. LSTM-AE and GRU-AE map the price sequence through 32 recurrent neurons to an 8-dimensional bottleneck.

Training is performed using the Adam optimizer with a learning rate of 10^{-3} , a batch size of 32, and 40 epochs. Training uses mean squared error loss [5], [11]. The GRU version includes the input and forget gates, which increase the training speed by 25% while maintaining temporal dependency. The DDPM model learns to reverse the process of adding Gaussian noise to data on discrete schedules [7]. During the inference stage, we corrupt window samples at four noise levels ($t \in \{0.10, 0.25, 0.40, 0.60\}$) 16 times and average the resulting residuals to obtain the anomaly scores.

To investigate the relevance of regional transmission context to prediction, we also input SP15, NP15, and

ZP26 together into two multi-hub models: a 3-channel multivariate LSTM control and a graph attention network (GNN-AE). In the GNN-AE model, the regional hubs are represented as a fully connected graph with three nodes, where the connections between nodes reflect the transmission ties for Path 15 and Path 26. This graph attention network utilizes two graph attention layers (with 8 hidden units), where the self-attention weight α_{ij} is computed using LeakyReLU on the concatenated node embeddings and is followed by LSTM-based temporal sequence encoding.

We conducted all experiments using five random seeds (0 to 4), ensuring that `torch.set_num_threads(1)` was always set and that `cuDNN` flags were fixed to ensure determinism.

V. Detection Results and Model Comparison

We optimized our thresholds on the validation set by varying the budget percentiles from 0.5% to 10% to maximize F_1 , then locked these parameters before testing on the test set.

Results for all models in terms of reconstruction quality and anomaly detection performance are summarized in Table II. The test ROC curve is illustrated in Fig. 2, while Fig. 3

Perhaps the most obvious trend from Table II is that low reconstruction error does not necessarily indicate high-quality anomaly detection. While the VAE yielded the highest-quality reconstructions out of all single-node models (RMSE \$7.85/MWh), it generated the lowest detection scores (AUROC = 0.534; $F_1 = 0.027$). The VAE relies on the latent prior to generate Gaussian noise, so the output becomes smoothed, eliminating the exact signals needed for anomaly detection.

The performance of recurrent autoencoders was much superior due to their ability to handle temporal depen-

TABLE II
Master Performance Summary: Reconstruction Fidelity and Anomaly Detection Across All Models

Model Architecture	Reconstruction Error ($\downarrow$)		Anomaly Detection Accuracy ($\uparrow$)				Operating Point Metrics ($\uparrow$)		
	RMSE (\$)	MAE (\$)	Mean AUROC	Ens. AUROC	Avg. Prec.	W2 AUROC	F_1 Score	Precision	Recall
Single-Node Models (Evaluated on CAISO SP15 Hub)									
iForest	n/a	n/a	0.623 ± 0.030	0.624	0.138	0.453	0.222 ± 0.028	0.227	0.217
VAE	7.85 ± 0.47	6.21 ± 0.39	0.534 ± 0.164	0.576	0.007	0.673	0.027 ± 0.021	0.015	0.226
Diffusion (DDPM)	n/a [†]	n/a [†]	0.564 ± 0.071	0.631	0.009	0.744	0.016 ± 0.015	0.009	0.096
LSTM-AE	10.37 ± 1.76	8.44 ± 1.45	0.738 ± 0.145	0.859	0.184	0.921	0.107 ± 0.099	0.073	0.261
GRU-AE	9.70 ± 0.68	7.77 ± 0.68	0.776 ± 0.237	0.882	0.162	0.894	0.157 ± 0.152	0.110	0.304
Multi-Node Spatial Models (Simultaneously Processing All 3 Regional Hubs: NP15, SP15, ZP26)									
MV-LSTM Control	9.12 ± 0.84	7.15 ± 0.72	0.854 ± 0.165	0.926	0.201	n/a	0.185 ± 0.092	0.142	0.326
GNN-AE	8.86 ± 0.65	6.94 ± 0.58	0.938 ± 0.098	0.985	0.246	n/a	0.246 ± 0.081	0.212	0.391

Notes: $\downarrow$ Lower error is preferred; $\uparrow$ Higher accuracy is preferred. [†]Diffusion scores are derived from normalized noise residuals; iForest lacks a decoder. Multi-node setups process SP15, NP15, and ZP26 simultaneously.

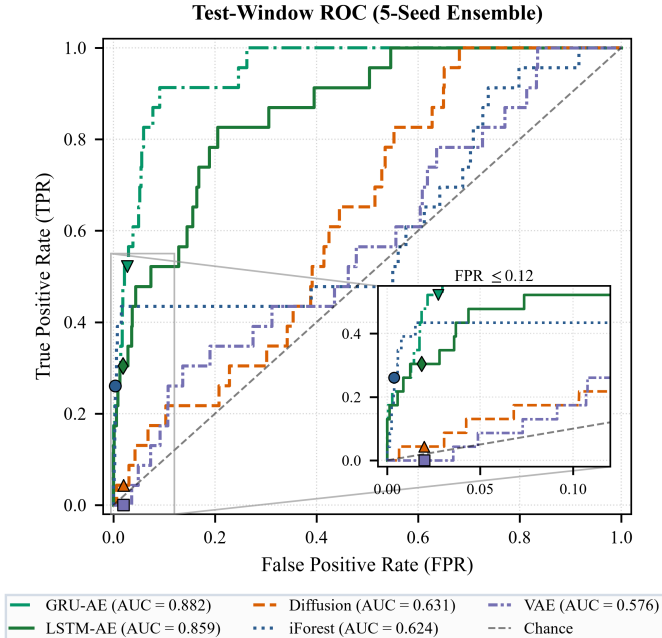


Fig. 2. Test-window ROC curves for the 5-seed ensemble models on CAISO data, GRU-AE (downward triangle), LSTM-AE (diamond), Diffusion (upward triangle), iForest (circle), VAE (square), with the inset showing the operational low-false-positive region ($FPR \leq 0.12$).

dencies. The AUROC values for the LSTM-AE and GRU-AE models were 0.859 and 0.882, respectively. As for the reason behind their superiority in this matter, one should note that, due to the sequential updating process, recurrent models maintain their memory of a typical daily pattern. Thus, upon encountering a sudden price change, the hidden state will be unable to reconcile the jump, and the error rate will skyrocket.

Incorporating price information from the three California centers increased the GNN-AE's mean AUROC to 0.938. However, the simple multivariate LSTM baseline performed nearly as well, achieving a mean AUROC of 0.854 and reaching 0.988 AUROC on three of the five seeds.

Comparison of the two with a paired bootstrap yielded a 95% confidence interval of $[-0.102, +0.281]$; as the interval includes zero, there is no statistically significant difference between using graph attention and using multivariate modeling alone. The added value seems to be mostly due to tracking price behavior in neighboring regions, rather than to graph attention.

The second test window for next year would demonstrate that the recurrent models can generalize across seasons, as the LSTM-AE achieved an AUROC of 0.921. On the other hand, the performance of the static Isolation Forest was extremely poor, as it could only get an AUROC of 0.453, worse than guessing randomly, because it is incapable of adapting to wholesale price variations from winter demand to summer solar/hydropower production.

The results obtained for a 5-channel diffusion model trained across all three hubs and two ancillary streams were an average AUROC of 0.528 ± 0.066 , which is lower than the default at one hub (0.564 ± 0.071). It seems like the additional price channels did nothing but add noise to the forward diffusion process, providing nothing to denoise on the reverse pass.

VI. Turning Detection into Profit

In order to examine whether the anomaly alerts result in profits, we conducted trading simulations for a battery that can trade with a capacity of 10 MWh/10 MW ($e = 0.80$, transaction cost \$0.50/MWh, degradation cost \$2.50 Strategy 1 (Heuristic dispatch) is activated every time prices are less than the 25th percentile of the historical value and ends as soon as the prices become greater than the 75th percentile, with 5 MW dispatch and 10 MW dispatch as the event occurs. Strategy 2 (Directional spread) goes long or short 1 MWh for every anomaly detected, with a 24-hour trading period. Strategy 3 (Tabular Q-learning) learns to dispatch in a discrete state space defined by $s = (p_z, a_{\text{flag}}, s_{\text{soc}})$. Discrete z-normalized price is $N_p = 10$ bins in the range from -3 to $+3$; anomaly flag is $N_a = 5$ categories (or $N_a = 1$

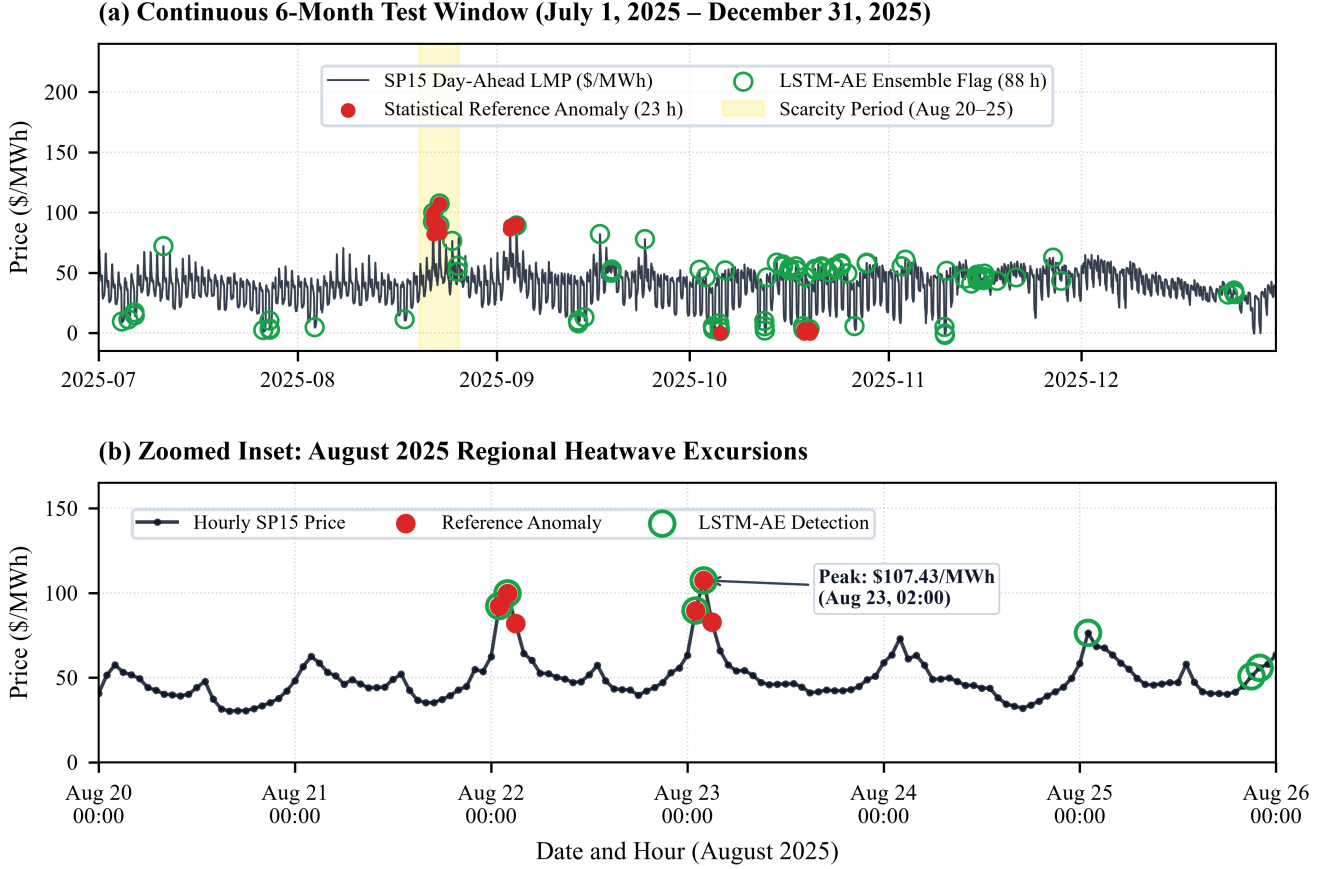


Fig. 3. Price spike identification for 6-month test data SP15 using reference (closed circles) and LSTM-AE (open squares) detection; it is clearly visible that the LSTM-AE detection agrees with the price spikes.

TABLE III

Arbitrage Earnings Across Both Test Windows (LSTM-AE Signal)

Trading Strategy	Window 1 Profit	Window 2 Profit
Heuristic (Without / With Signal)	\$13,085 / \$13,089	\$30,197 / \$30,762
Directional Spread (Trades, Win%)	\$487 (68, 76%)	\$565 (84, n/a)
Q-Learning: Without Signal	\$3,090 $\pm$ \$3,905	\$23,451 $\pm$ \$3,180
Q-Learning: With Signal	\$11,957 $\pm$ \$393	\$35,567 $\pm$ \$3,658
Deep Q-Network (DQN) (Without / With)	\$5,931 / \$5,246	n/a

for unconditional version); battery state-of-charge ratio ($SOCh_t/C$) is $N_s = 5$ bins resulting in $|S| = 250$ discrete states ($|S| = 50$ without anomaly flag). The action space contains three discrete power actions $a \in \{-1.0, 0.0, 1.0\}$ (discharge, do nothing, charge). The action values are iteratively learned in 30 training runs using Eq. (2):

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r_t + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (2)$$

with learning rate $\alpha = 0.15$, discount factor $\gamma = 0.95$, and an ϵ -greedy exploration schedule decaying linearly from $\epsilon_{\text{start}} = 0.30$ to $\epsilon_{\text{end}} = 0.02$.

According to Table III and Fig. 4, the profits from trading depend greatly on the degree of flexibility of the controller. In case of heuristic dispatching, the anomalous signal had almost no effect on profits (+0.03% for Window 1, +1.87% for Window 2) because power ratings and SoC constraints restrict the actions of static-threshold rules.

By contrast, directional spread trading revealed that the signal had value, as it delivered \$487 in profit with a 76% success ratio, whereas under pure randomness it yielded only \$30.

The largest improvement was observed in Tabular Q-learning. The possibility for an agent to learn to use anomaly signals led to a near fourfold increase in Window 1 profit from \$3,090 to \$11,957 and a decrease in cross-seed standard deviation from \$3,905 to \$393 (-90%). In addition, Window 2's profit increased by 51.7% to \$35,567.

Nevertheless, Compute matched DQN gained \$5,246 profit using the signal, but the no-signal situation brought \$5,931 profit. Hence, continuous deep reinforcement learning was unable to handle reward noise in time series, whereas tabular Q-learning was able to generalize regardless of such noise. Please see Table IV for the detailed results for each detector in Window 1, where LSTM-AE turned out to be the most efficient with 76% win ratio and \$487 in spread profit.

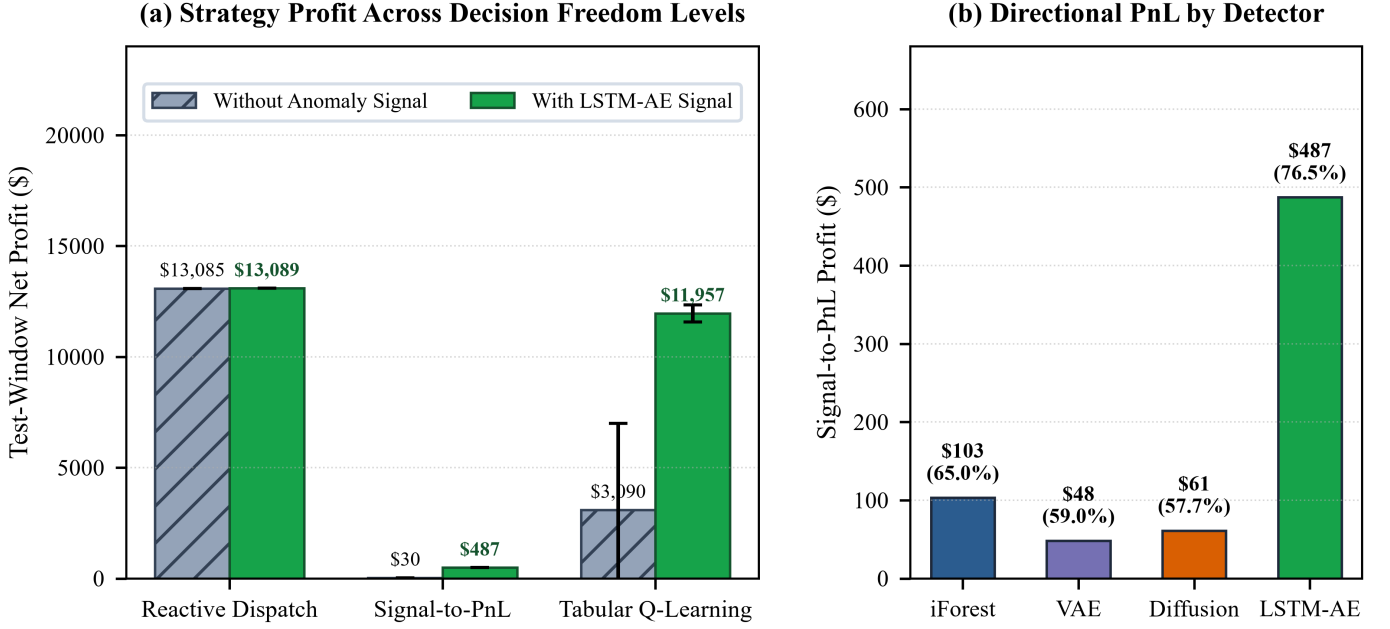


Fig. 4. Window 1 cumulative trading returns. (a) Heuristic dispatch, directional spread trading, and tabular Q-learning, showing how the anomaly signal nearly quadruples Q-learning profit while cutting seed variance tenfold. (b) Directional arbitrage profit and win rate across the four detector families.

TABLE IV
Per-Detector Arbitrage Performance (Window 1)

Detector Family	Dispatch (\$)	Spread PnL (\$)	Trades, Win%
iForest	13,128	103	20, 68%
VAE	13,017	537	220, 61%
Diffusion	12,740	60	13, 77%
LSTM-AE	13,089	487	68, 76%

TABLE V
Adversarial Detection and Economic Recovery (5 Seeds)

Attack Profile	Recall	AUROC	Damage (%)	Recovery (%) (F_1 / Cons. / Oracle)
Naive	0.76 ± 0.07	0.973 ± 0.026	-7.18 ± 0.92	59.2 / 33.8 / 89.1
Adaptive	0.66 ± 0.03	0.971 ± 0.011	-5.10 ± 0.89	36.5 / 19.8 / 83.3
Sinusoidal	0.75 ± 0.02	0.980 ± 0.011	-15.11 ± 0.91	67.2 / 66.2 / 94.0

VII. Adversarial Robustness: A Companion Study

We also considered the reverse scenario in this problem, where an attacker would inject fake price spikes into the settlement feed system. In such a situation, the battery discharges energy into the fake price spikes, making it appear to make money, even though it does not receive any payment.

Three types of attack profiles were executed on the SP15 price: the naive multiplicative $3.0\times$ attacks per hour, an 8-hour raised-cosine price attack intended to replicate the morning ramp attack profile, and multi-hour sinusoidal price wave attacks. The defense strategy involves the XGBoost predictor using 17 causal variables: z-score, trailing ratio, volatility, and consensus. If an hour is attacked, the hour's trailing prediction will be used.

In Table V, we see how much money is being lost without our defender: -7.18% for naive spikes, -5.10% for adaptive bumps, and -15.11% for sinusoidal waves. With our filter, we can recover the vast majority, 59.2%, 36.5%, and 67.2%, of those losses, respectively, and come very close to our theoretical oracle bounds of 83.3% to 94.0%.

TABLE VI
Break-Even Precision: Machine-Learning Classifier vs. Simple Threshold

Defense Strategy	Prec.	\$/TP Saved	\$/FP Lost	p^* Break-Even
Learned Classifier	0.891	\$24.09	-\$19.11	0.334
Simple Robust-z Rule	0.802	-\$4.89	-\$350.45	1.017

Notes: Reported p^* is the cross-seed/attack mean ($\mathbb{E}[p_i^*]$ across runs) rather than a ratio of aggregate column averages.

A. Break-Even Precision Analysis (p^*)

In the battery dispatch problem, a true positive (TP) identifies a fake spike, thus earning a profit ($v_{TP} > 0$). In contrast, a false positive (FP) misidentifies a genuine price spike, resulting in revenue loss ($v_{FP} < 0$). An operational defense that operates at a precision level of p is said to be at the break-even point if the expected value of the cash flows:

$$p^* = \frac{v_{FP}}{v_{FP} - v_{TP}} = \frac{|v_{FP}|}{|v_{FP}| + v_{TP}}. \quad (3)$$

Economic differences between the learned classifier and the threshold-based classifier are quite remarkable (Table VI and Figure 5). The learned classifier is estimated

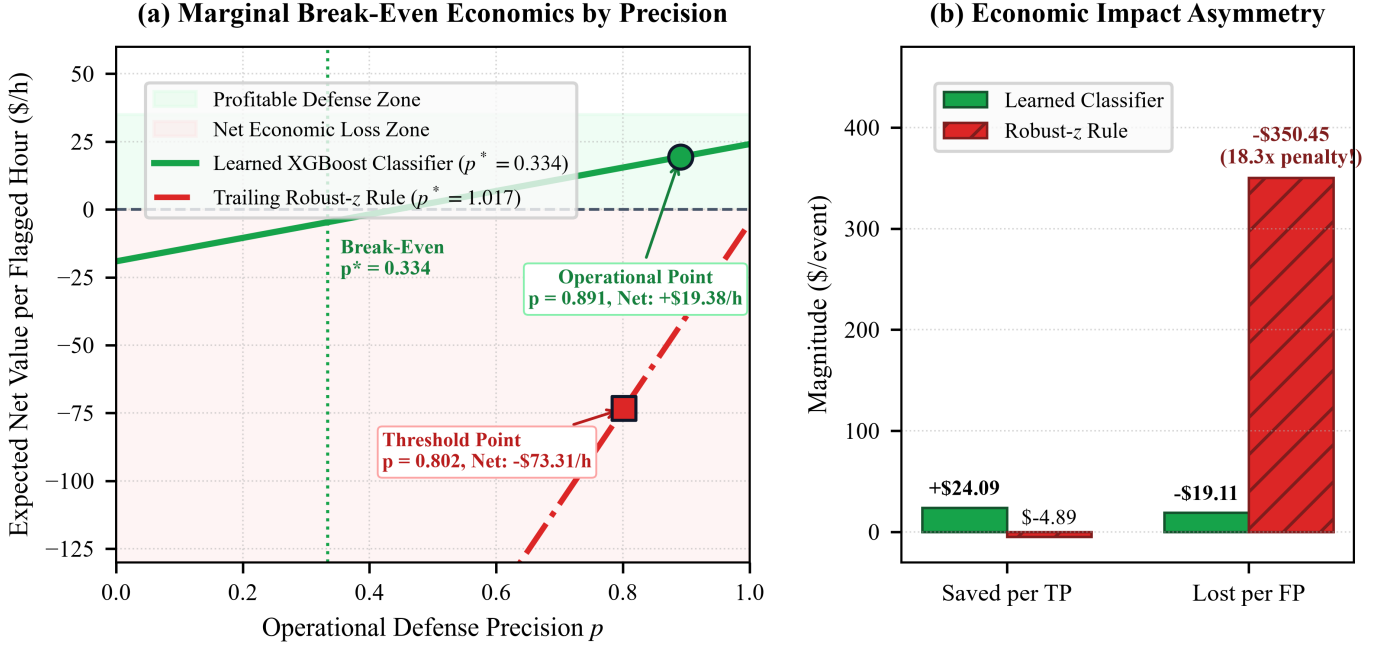


Fig. 5. Break-even precision trade-off. (a) Net cash flow versus precision p for the learned classifier ($p^* = 0.334$, operating at $p = 0.891$) against the simple threshold rule ($p^* = 1.017$). (b) The asymmetry in false-alarm penalties: a threshold false alarm costs $18.3\times$ as much as a true positive saves.

to save, on average, \$24.09 for every detected price spike and spend \$19.11 on each false alarm; hence, the break-even point is reached at the precision level of $p^* = 0.334$. Given that the learned classifier’s actual precision is 89.1%, it generates profit with ease.

On the other hand, a robust- z threshold-based classifier is significantly less effective in terms of economic performance. This classifier incurs, on average, \$350.45 for each false alarm, approximately 18.3 times higher than for the learned classifier. Because high absolute values trigger the rule, it misses the year’s most valuable price spikes; hence, its break-even point is $p^* = 1.017$, over 100%.

B. Generalization for Cross-Attacks

From the cross-attacks experiments (Table VII and Figure 6), it becomes clear that defenders need to learn their defenses on different threat models. Nominal AUROC score stays above 0.75, while economic gain drops in the case when the threat model used in the attack is not known to the defender. For example, there was a decrease in economic gain by 40.9% for a model that learned to defend from sinusoidal waves against adaptive bump attacks.

VIII. Discussion and Practical Takeaways

The results can be applied in practice in three ways to storage in wholesale electricity markets.

Why do recurrent models emerge victorious in this case? Electricity prices are not independent samples from some distribution; they depend on the solar cycle, industrial activity, and the speed at which thermal power stations come online. Hence, when there is an unusual jump of \$800, an LSTM or GRU does not reconstruct this jump

correctly because its internal state has been influenced by the previous 20 hours of normal diurnal dynamics, something which is impossible for a static model or a VAE, which treats all the hours equally.

Another clear implication is that anomaly-based solutions can work only when the defender has access to the alarm. Some algorithms would find it hard to apply the alarms because their deployment was based on the thresholds anyway. Tabular Q-learning, on the other hand, used this data to reduce energy usage during normal times and sell the saved energy at peak times.

The charts clearly illustrate another key concept related to grid security: false positives during real price spikes are simply too expensive. In order to make up for a loss of \$350 because of just one false positive, one needs to detect hundreds of attacks correctly; therefore, any solution used in practice should preserve real spikes at first.

IX. Limitations and Future Work

The 4-MAD reference flag is an objective benchmark in statistical scoring for detection, but actual trading gains are what matter. Furthermore, our dispatch optimization problem follows a linear degradation cost function of \$2.50 per MWh in the algorithm performance measures; incorporating the more realistic, non-linear rainflow count with electrochemical cell heat dynamics will improve long-term profits. In our reinforcement learning experiment, we observe that tabular Q-learning is robust, whereas DQN-based approximations in continuous spaces struggle with market disruptions. Lastly, we have experimented with the algorithm using only three price regions for spatial

TABLE VII
Cross-Attack Transfer Performance Across Source and Target Threat Types (5-Seed Average)

Source Training Attack	Target Test Attack	AUROC	Precision	Recall	Recovery %
Naive	Naive	0.973	0.979	0.721	33.8
Naive	Adaptive	0.947	0.989	0.476	-18.0
Naive	Sinusoidal	0.907	0.994	0.224	5.2
Adaptive	Naive	0.973	0.967	0.697	50.6
Adaptive	Adaptive	0.971	0.962	0.737	19.8
Adaptive	Sinusoidal	0.885	0.952	0.266	8.6
Sinusoidal	Naive	0.847	0.753	0.353	18.7
Sinusoidal	Adaptive	0.753	0.717	0.114	-40.9
Sinusoidal	Sinusoidal	0.980	0.963	0.779	66.2
Adaptive + Sinusoidal	Naive	0.960	0.881	0.674	43.2
Naive + Sinusoidal	Adaptive	0.937	0.995	0.352	-46.6
Naive + Adaptive	Sinusoidal	0.902	1.000	0.200	3.9

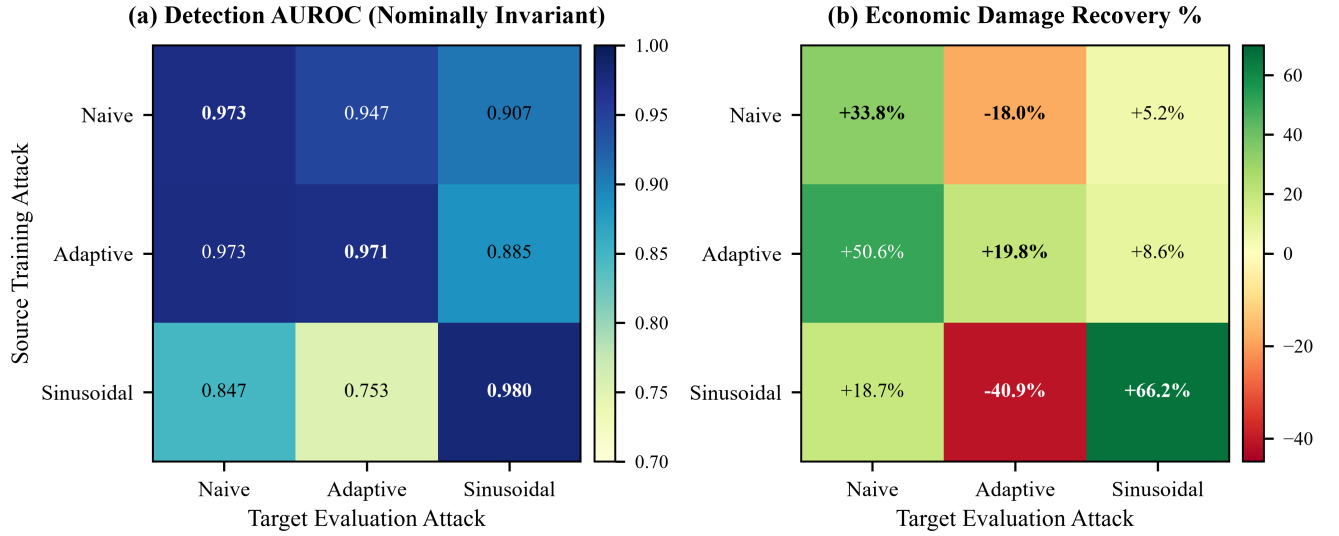


Fig. 6. Cross-attack transfer evaluation. (a) The detection AUROC matrix indicates uniformly high nominal values (≥ 0.75). (b) The economic recovery matrix illustrates serious cross-attack weaknesses; for instance, -40.9% when moving from Sinusoidal to Adaptive attacks.

analysis, while a key future step is to experiment with graph neural networks on high-transmission topologies.

X. Conclusion

In this study, we examine the problem of price anomaly detection in battery energy storage systems using three years of raw, real data from the CAISO wholesale electricity market. It was shown that recurrent autoencoders were the best-performing architecture for price anomaly detection in practice, yielding AUROC scores of 0.859 for LSTM-AE and 0.882 for GRU-AE. By feeding the output of the anomaly detection model to a tabular Q-learning dispatch controller, the profit of a battery was improved by \$8,867, and the variability of profits was decreased by a factor of 10. In our adversarial experiments, we have revealed the inefficiency of threshold-based rules by showing that their break-even point in precision lies far beyond $p^* > 1.0$, implying that these methods block the most profitable spikes in a market. On the contrary, a trained classifier needs only 33.4% precision to break even,

operates at 89.1%, and recovers up to 67.2% of attack losses.

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